

Xinbao Qiao

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EDUCATION

The Chinese University of Hong Kong Hong Kong, China
PhD candidate in Information Engineering 2026–present

Zhejiang University Hangzhou, China
M.Sc. in Artificial Intelligence 2022–2025

- Major GPA: 90/100; rank: 3/25.
- Selected coursework: AI Algorithms and Systems (100), Secure Artificial Intelligence (97).

Shandong University Jinan, China
B.Eng. in Communication Engineering 2018–2022

- Major GPA: 82.47/100.
- Third-class Academic Award, 2018–2019.

RESEARCH INTERESTS

AI and networks; distributed Wasserstein barycenter computation; trustworthy AI; data-centric machine learning; machine unlearning; synthetic-data evaluation and model collapse.

PUBLICATIONS

- [1] **When Sample Selection Bias Precipitates Model Collapse.**
Xinbao Qiao, Xianglong Du, Wei Liu, Jingqi Zhang, Peihua Mai, Meng Zhang, Yan Pang. *ICML*, 2026.
- [2] **Beyond Binary Erasure: Soft-Weighted Unlearning for Fairness and Robustness.**
Xinbao Qiao, Ningning Ding, Yushi Cheng, Meng Zhang. *AAAI*, 2026.
- [3] **Hessian-Free Online Certified Unlearning.**
Xinbao Qiao, Meng Zhang, Ming Tang, Ermin Wei. *ICLR*, 2025.
- [4] **DynFrS: An Efficient Framework for Machine Unlearning in Random Forest.**
Shurong Wang, Zhuoyang Shen, Xinbao Qiao, Tongning Zhang, Meng Zhang. *ICLR*, 2025.
- [5] **Learn What Matters: Data Pruning for Efficient Decentralized Learning.**
Xinbao Qiao, Xunhao Jiang, Zuozhu Liu, Peng Sun, Meng Zhang. Under review.

RESEARCH EXPERIENCE

The Chinese University of Hong Kong Hong Kong, China
PhD candidate and Research Assistant 2026–present

- Study AI and networks, with emphasis on distributed computation, communication-aware learning, and reliable evaluation under decentralized data access.
- Current focus: distributed Wasserstein barycenter computation for collaborative distributional references across local empirical measures.

National University of Singapore Research Institute (Chongqing) Chongqing, China
Full-time Research Intern; advisor: Yan Pang 2025

- Worked on trustworthy LLM systems and synthetic-data evaluation.
- Developed collaborative Wasserstein-geometry methods for evaluating recursive synthetic-data training under local sample-selection bias.

Zhejiang University Hangzhou, China
M.Sc. researcher; advisor: Meng Zhang 2023–2025

- Studied data influence attribution, certified data removal, and machine unlearning.
- Developed methods that examine trade-offs among fairness, robustness, privacy, and utility in data-centric learning systems.

SELECTED ACADEMIC PROJECTS

- **AI and networks:** study distributed computation and collaborative evaluation for learning systems whose data and evidence are spread across networked parties.
- **Synthetic-data model collapse:** analyzed how locally biased sample selection can accelerate model collapse, and proposed collaborative Wasserstein proxy references for distributed evaluation.
- **Machine unlearning:** designed Hessian-free certified unlearning and soft-weighted unlearning methods for privacy, fairness, and robustness settings.

SKILLS

Programming: Python, PyTorch, TensorFlow, MATLAB, C/C++, SQL, Linux, Docker, Git, L^AT_EX.

Languages: Mandarin Chinese (native), English.